



Waipapa
Taumata Rau
University
of Auckland

An introduction to systems thinking in ENGGEN403

Lecture 2

➤➤➤ July 26

➤➤➤ Amanda di Ienno and Juliet Gerrard



Outline

- Course schedule, CANVAS guide, assessments
- What is systems thinking?



Learning Outcomes

At the end of this session, students should be able to:

- Understand the course contents and format for the semester
- Describe the basics of Systems Thinking

Home

Announcements

Syllabus

Panopto Video

Modules

Ed Discussion

Assignments

Quizzes

Grades

SET Reports

SET Evaluations



Getting Started
| Hei Timata



Meet the Team
| Ngā Poutoko



Course Schedule
| Hōtaka Ako



Assessments |
Ngā Aromatawai



Support &
FAQ | Pātaitai



Syllabus | Marautanga



Contact form



Team Project
Resources
(Available Week 6)



Systems Week
Resources
(Available Week 9)

Week 1 begins 20 Jul	• What can ENGGEN 403 do for me? • Introduction to Systems Thinking in ENGGEN 403 • Interface of Science, Engineering, & Government - some real world examples	
Week 2 begins 27 Jul	• Business Case Analysis and Option Assessment for System-Level Change • Social Cost-Benefit Analysis – Valuing what matters • GDP, Government spending, and things that metrics miss	• Individual Goal Setting (8%)
Week 3 begins 3 Aug	• Stakeholders at the national level • Systems thinking and wicked problems • Defining and scoping your wicked problem	• Quiz 1 (5%)
Week 4 begins 10 Aug	• Machinery of Government • Guest lecturer: Tom Wilson, Chief Science Advisor to the National Emergency Management Agency • Pulling it all Together: what have we learned?	• Guest Lecture Participation (1%)
Week 5 begins 17 Aug	• Some worked examples ahead of your assessments • Getting your team match fit: Team Canvas Workshop • The big reveal: Team Project Brief	• Quiz 2 (5%) • Team Match Fit Lecture Participation (Team Canvas) (1%)
Week 6 begins 24 Aug	• Team Project	• Team Project (25%)
Mid-Sem Break begins 31 Aug	Mid-Semester Break	• Team Project Peer Review 1 (1%)

Week 7 begins 14 Sep	• McKinsey workshop • Teamwork, Team Debrief, Managing Teams in Systems Week • Team project feedback – what have you learned ahead of Systems Week?	
Week 8 begins 21 Sep	• Timelines for wicked problems • Graduate Panel on Systems Week (Live Zoom webinar) • The really big reveal: Systems Project Brief	
Week 9 begins 28 Sep	• SYSTEMS WEEK	• Systems Project (50%)
Week 10 begins 5 Oct	• No Lectures	• Systems Project Peer Review 2 (1%)
Week 11 begins 12 Oct	• Systems week feedback (Asynchronous Lecture)	
Week 12 begins 19 Oct	• No Lectures	• Reflection (3%)

- Week 6: the Team project is a “practice run” for Systems Week - Formative Assessment
- Takes place Monday 8 am to Friday 5 pm – supported by teaching staff clinic,
- No ENGGEN 403 Lectures that week, the clinic runs 10 am – 4 pm each day
- Teams may work remotely
- Alongside your other courses



- You are a large team of consultants working a very full week on a real world national level systems problem to support the NZ government
- **All other part IV engineering course lectures cancelled**
- **In-Person sessions compulsory**
- **Online sessions and other resources available**
- Teaching staff clinic 9-5 each day
- Everyone must contribute



Quizzes

Thursday 9 am to Friday 10 pm

- 2 quizzes: Wk 3 & 5
- 60 minutes to complete
- Make sure to read the quiz resource pages

Lecture Participation: wk 4 & 5

- Guest Lecture
- Organising the Team

Team Peer Review: MSB & wk 10

- Rate and provide feedback to your team



Individual Goal Setting

Two steps

- 1) Reflect & Set SMART goals
- 2) Peer review others' reflections and goals

Step 1: Opened July 21 at 2pm
Due Monday 27 July at 10 pm
Lecture 1 may be helpful
Must submit step 1 to do step 2

Step 2: Opens Wednesday July 29 at 12 pm
Due Tuesday August 4 at 10 pm



Personal Reflection

Reflect on your experience of
ENGGEN 403

Did you complete your goals?

What learnings will you take with
you?

We read every one of these!



Why academic integrity is important:

- Academic integrity is the honesty that you have in your studies and assessments. It is a key foundation to being a member of the University's academic community.
- Having academic integrity shows that you respect the work of other people, such as their Ideas, Words, Image, Code
- Breaching academic integrity leads to penalties and consequences, e.g. damaged reputation.

NEW ZEALAND

Engineer suspended after signing off on unsafe tuk-tuks



NZ Herald

14 Jan, 2020 02:53 PM 2 mins to read

Save Share



A Tuk-tuk signed off as safe by the engineer rolled in Wellington in 2016. Photo / 123RF

- Academic misconduct may happen in coursework and exams.
- Examples: inadvertent plagiarism, collusion and contract cheating.
- Revisit the online Academic Integrity Course, ACADINT A01.
- Read more information about academic integrity.



... / [Student policies and guidelines](#) / [Academic integrity and copyright](#) / [About academic integrity](#) +

About academic integrity

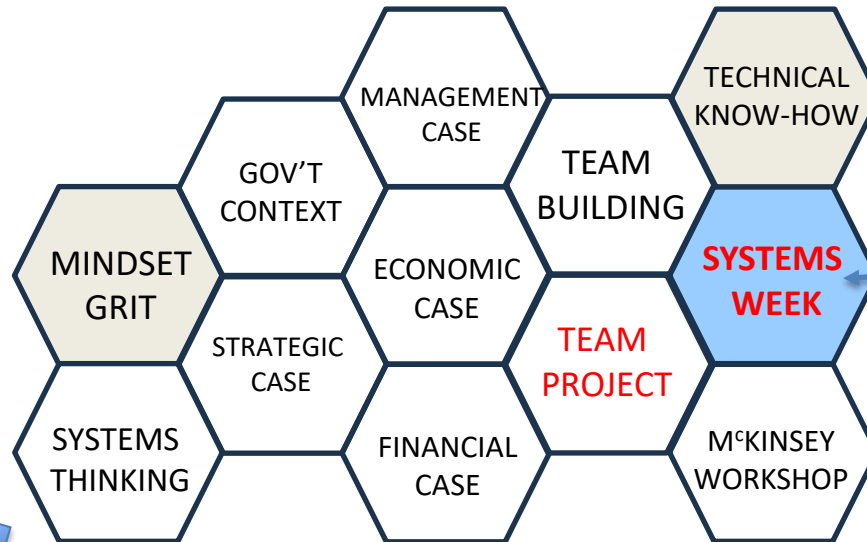
Find out what academic integrity is and why it's important for your studies.

- If you're unsure you're meeting academic integrity, don't be afraid to ask for help from lecturers and/or course coordinator.



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Controlled Assessment (Lane 1)	Uncontrolled Assessment (Lane 2)
AI is restricted.	AI may be used to assist student learning
Lecture participation	Everything Else
	For Team Project & Systems Week: Team needs to agree how to use AI. Team needs to complete an Reflection of AI Usage



Crazy sounding assessment
(until you hit the real world)

Wicked problems: unsolvable, undefinable, unverifiable, unique, uncertain

The term wicked problem is used in relation to issues that resist a complete or simple solution.

It is likely that wicked problems can be ameliorated but not solved, and that amelioration measures may create new problems.

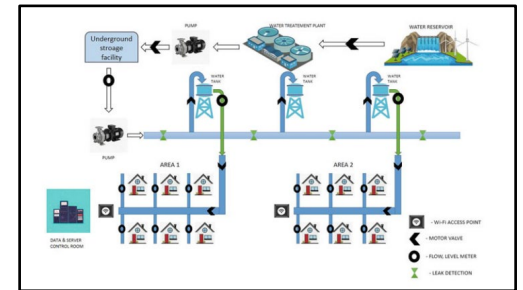
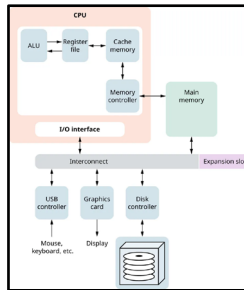
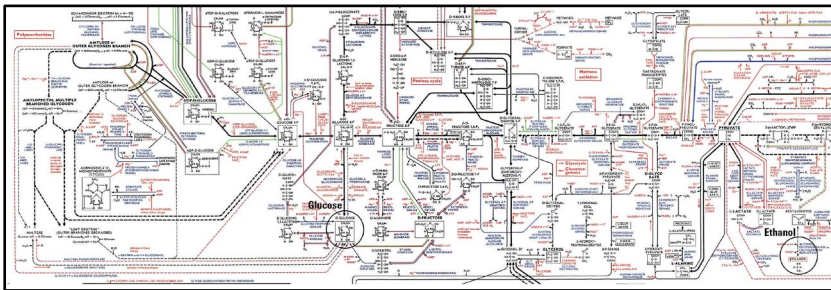
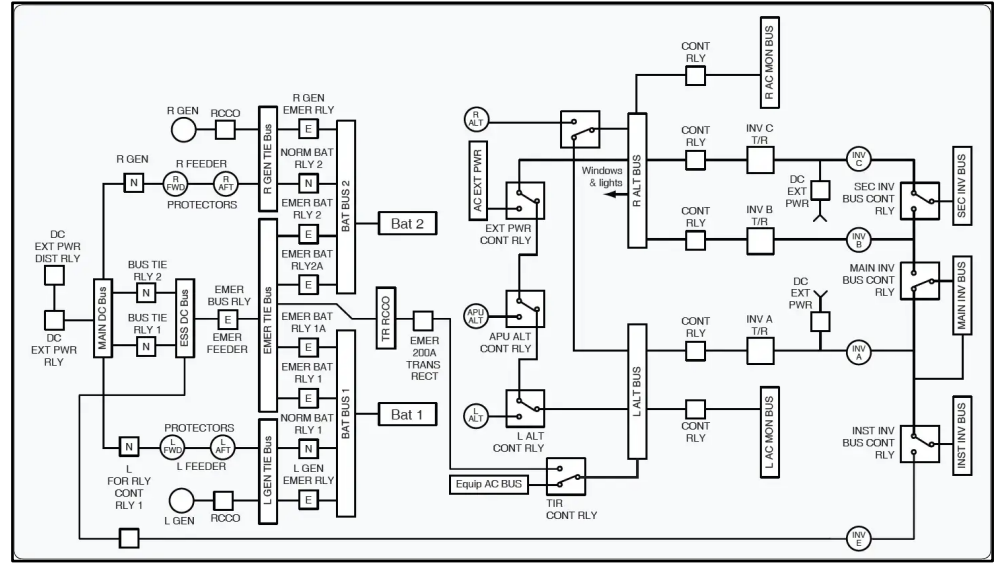
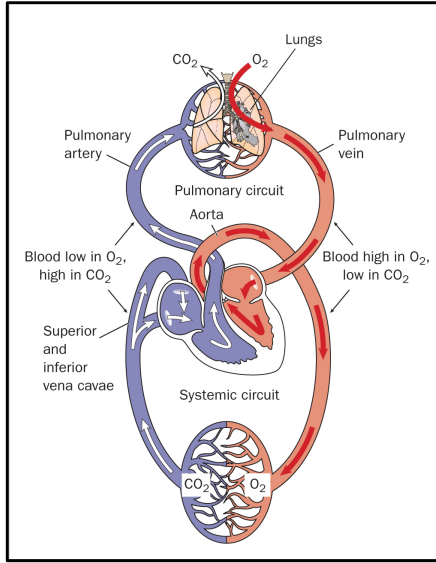
A wicked problem is distinct from a complex problem, which is hard, but may have an optimal solution, or a long problem, defined by its timeframe, although many wicked problems are often both complex and long.

Systems thinking: helps us ameliorate wicked problems



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'System' means different things to different people



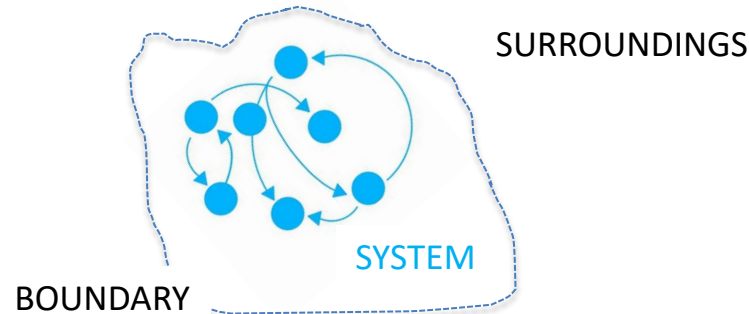
- In ENGGEN303, the system that you thought about tended to be a business and their stakeholders, usually customers
- This year, we are zooming out to look at big, gnarly, wicked problems that requires a wider systems view
- Your stakeholders now include the general public, local and central government, businesses, non-government organisations (NGOs), iwi, etc, etc
- It is likely that they all want different and potentially contradictory things
- Systems thinking will help you to navigate through the complexity and the tensions

- It is likely that your problem will have unclear definition and scope, complex stakeholders and dynamic power levels
- Relationships, power dynamics, interconnections and flows will all be important
- Solving problems like this requires collaboration across disciplines – as engineers you will have a useful skill set, but not the only skill set

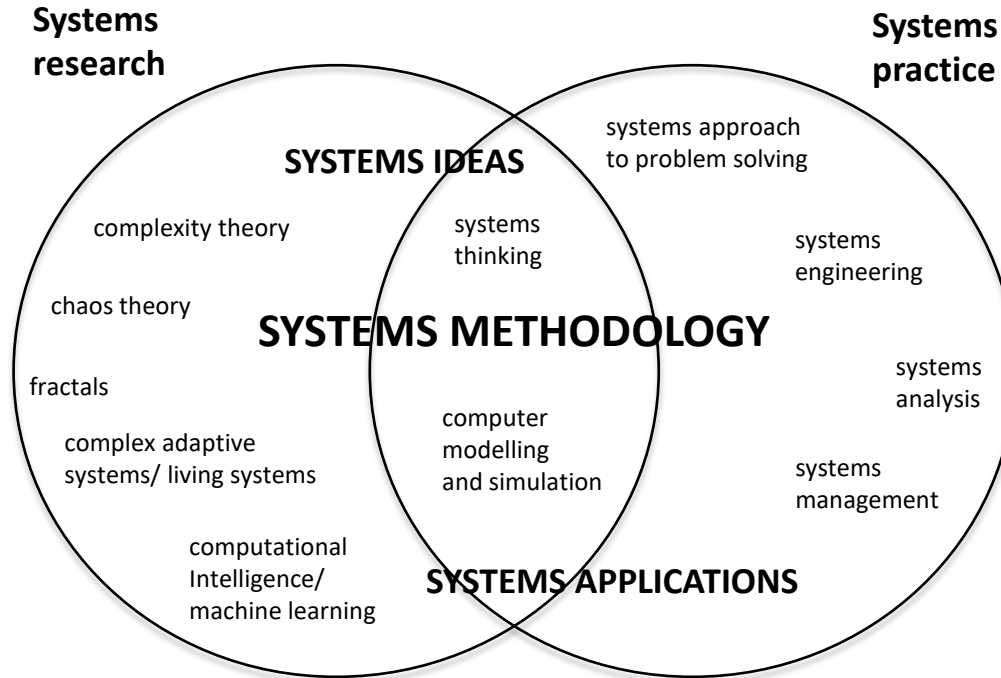


Image: Andrew Henning, Better Systems

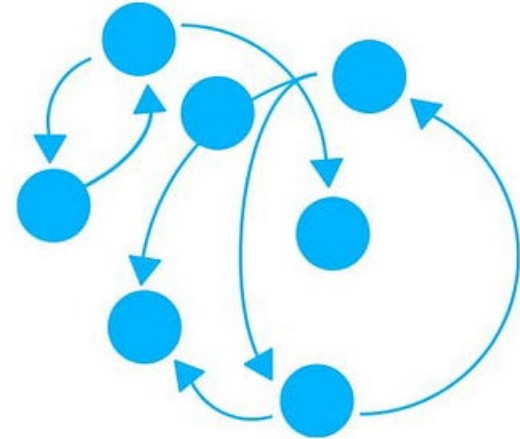
- **Example definition 1:** A system is a set of objects together with relationships between the objects and between their attributes. [...] For a given system, the environment is the set of all objects a change in whose attributes affect the system and also those objects whose attributes are changed by the behaviour of the system. (Hall & Fagan, 1956, pp. 18–20).
- **Example definition 2:** A set of connected parts that together form a cohesive whole

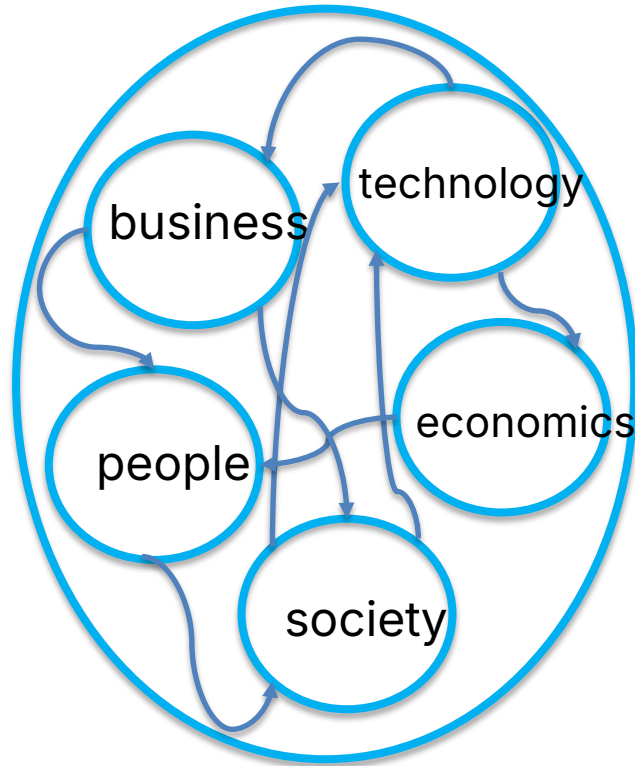


Systems thinking is a holistic and expansive approach to looking at situations and problems

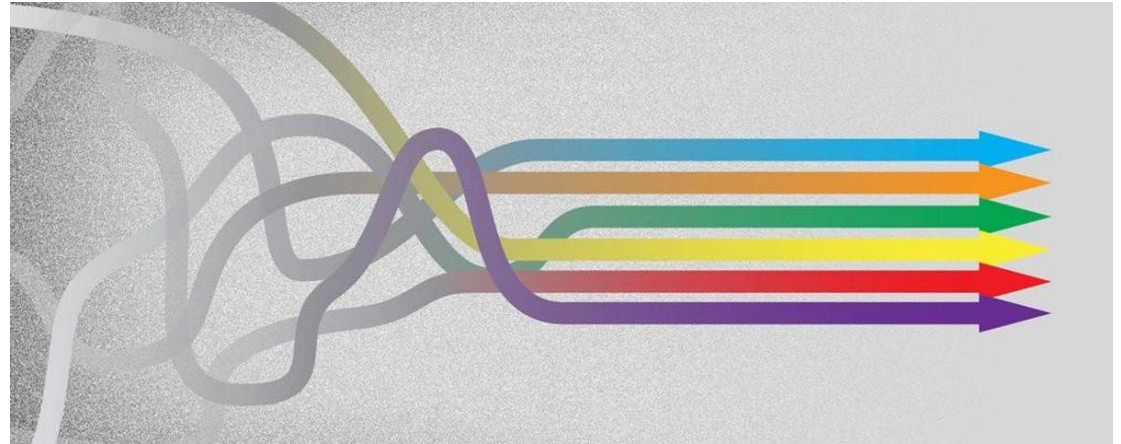


- Systems thinking is a holistic way to investigate factors and interactions that could contribute to a possible outcome. A mindset more than a prescribed practice, systems thinking provides an understanding of how individuals can work together in different types of teams and through that understanding, create the best possible processes to accomplish just about anything. (Morganelli, 2024)
- The discipline of systems thinking is more than just a collection of tools and methods – it's also an underlying philosophy. - Michael Goodman Systems Change Institute, Fifth Discipline Fieldbook (2004)



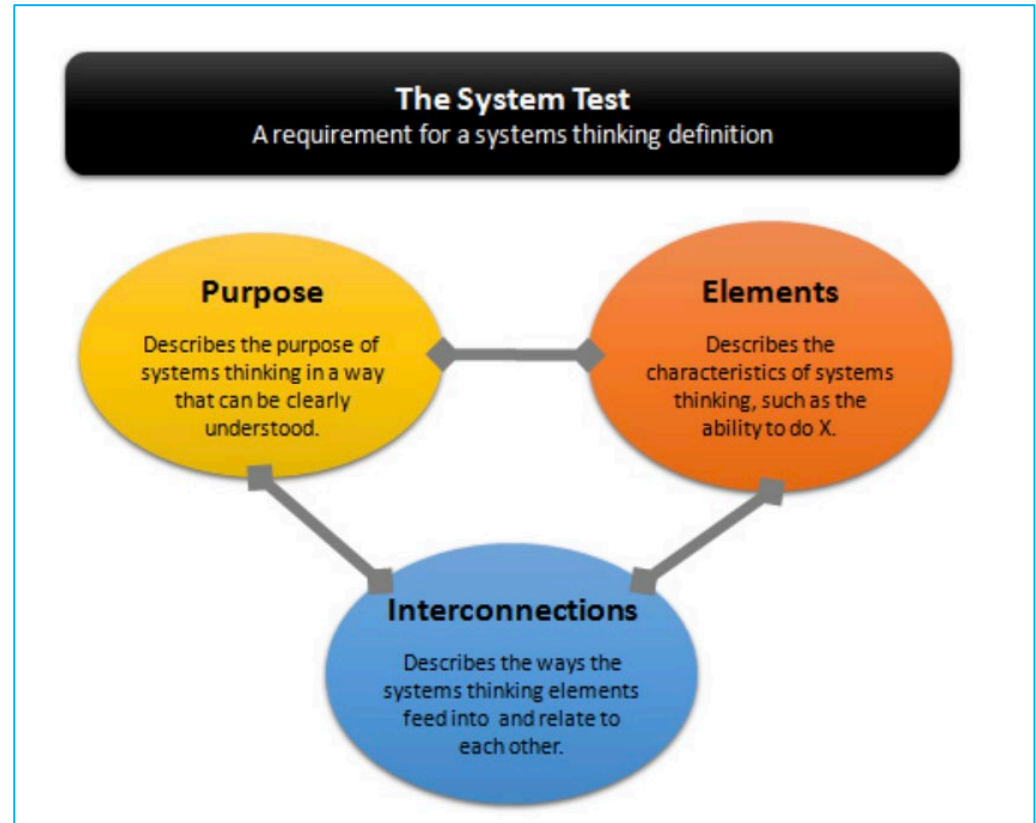


A process that uses a diverse group along with systems thinking tools to get from initial chaos to an agreed preliminary definition of the problem we are trying to resolve, and that then takes steps to develop an improvement or "solution."

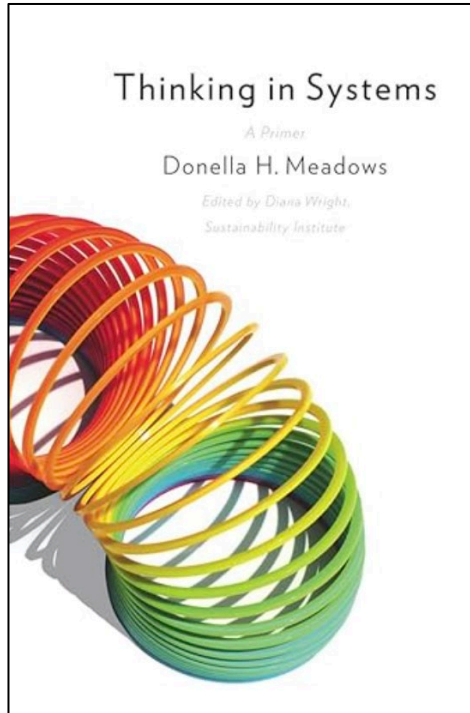


<https://blog.civismakers.com/strategic-alignment-the-key-to-organizational-success-b7b69fef911c>

Systems thinking is a set of synergistic analytic skills used to improve the capability of identifying and understanding systems, predicting their behaviours, and devising modifications to them in order to produce desired effects. These skills work together as a system, Arnold and Wade, 2015



Ahead of lecture 8: (at least the) first 2 chapters of *Thinking In Systems*



Reading goals:

- Hone your abilities to recognise systems and their parts
- Learn to look for interconnections
- Learn to ask "what if" questions about possible future behaviours
- Be creative and courageous
- Think about system design, and re-design



Questions?

Thank you, and see you next
lecture 😊

